

[1.] To find the generators of the Lorentz group that leave k^μ invariant, we have the following condition on the infinitesimal transformation of k^μ

$$0 \stackrel{!}{=} \delta k_\mu = \omega_{\mu\nu} k^\nu \implies \omega_{i0} + \omega_{i3} = 0 \text{ and } \omega_{03} = 0 = \omega_{30}.$$

Using these conditions we obtain

$$\begin{aligned} \omega_{\mu\nu} &= \frac{i}{2} \omega_{\alpha\beta} (M_{\mu\nu})^{\alpha\beta} \\ &= i\omega_{01} (M_{\mu\nu})^{01} + i\omega_{02} (M_{\mu\nu})^{02} + i\omega_{03} (M_{\mu\nu})^{03} + i\omega_{12} (M_{\mu\nu})^{12} + i\omega_{31} (M_{\mu\nu})^{31} + i\omega_{23} (M_{\mu\nu})^{23} \\ &= i\omega_{12} (M_{\mu\nu})^{12} + i\omega_{23} [(M_{\mu\nu})^{23} - (M_{\mu\nu})^{20}] + i\omega_{13} [(M_{\mu\nu})^{13} - (M_{\mu\nu})^{10}] \\ &= -i\omega_{12} (J^3)_{\mu\nu} - i\omega_{23} [(J^1)_{\mu\nu} + (N^2)_{\mu\nu}] - i\omega_{13} [-(J^2)_{\mu\nu} + (N^1)_{\mu\nu}]. \end{aligned}$$

Here, we can clearly see that the group which leaves k^μ invariant is three-parametric with the following generators:

$$J^3, \quad T^1 = J^1 + N^2, \quad T^2 = -J^2 + N^1.$$

Using the standard commutation relations of J^i, N^i generators

$$[J^i, J^j] = i\varepsilon^{ij}_k J^k, \quad [N^i, N^j] = -i\varepsilon^{ij}_k J^k, \quad [J^i, N^j] = i\varepsilon^{ij}_k N^k,$$

we obtain the commutation relations of the little group

$$\begin{aligned} [J^3, T^1] &= [J^3, J^1 + N^2] = iJ^2 - iN^1 = -iT^2 \\ [J^3, T^2] &= [J^3, -J^2 + N^1] = iJ^1 + iN^1 = iT^1 \\ [T^1, T^2] &= [J^1 + N^2, -J^2 + N^1] = -iJ^3 + iJ^3 = 0, \end{aligned}$$

which satisfy the standard commutation relations of the $\mathfrak{so}(2) \ltimes \mathbb{R}^2$ group.

[2.] We need to solve the equation

$$D(J^3) \cdot \varepsilon_\pm = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \pm \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix},$$

from which we immediately get

$$\varepsilon_\pm^\mu = \mathcal{N}_\pm (\mathbf{e}_1 \pm i\mathbf{e}_2).$$

The scalar products of these vectors are

$$\begin{aligned}\varepsilon_+ \cdot \varepsilon_+ &= -\mathcal{N}_+^2 (1 - 1) = 0 \\ \varepsilon_- \cdot \varepsilon_- &= -\mathcal{N}_-^2 (1 - 1) = 0 \\ \varepsilon_+ \cdot \varepsilon_- &= -\mathcal{N}_+ \mathcal{N}_- (1 + 1) = -2\mathcal{N}_+ \mathcal{N}_-\end{aligned}$$

To obtain the desired normalization

$$\varepsilon_+ \cdot \varepsilon_- = -1,$$

we symmetrically choose $\mathcal{N}_+ = -\mathcal{N}_- = \frac{1}{\sqrt{2}}$.

[3.] For the T^i generators, we have

$$\begin{aligned}D(T^1) \cdot \varepsilon_{\pm} &= \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix} \frac{\mp 1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix} = \frac{-1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \neq 0 \\ D(T^2) \cdot \varepsilon_{\pm} &= \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & -i \\ 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix} \frac{\mp 1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix} = \frac{\mp 1}{\sqrt{2}} \begin{pmatrix} i \\ 0 \\ 0 \\ i \end{pmatrix} \neq 0\end{aligned}$$

[5.] Using the hint, we have

$$\varepsilon_{\pm}(\Lambda p) = \Lambda L(p) W^{-1}(\Lambda, p) \varepsilon_{\pm}(k) = \Lambda L(p) e^{-i\vartheta(p, \Lambda) J^3} e^{-ib^j(p, \Lambda) T_j} \varepsilon_{\pm}(k).$$

Now we use the results of the 2nd and 3rd exercise. We know that

$$J^3 \cdot \varepsilon_{\pm}(k) = \pm \varepsilon_{\pm}(k)$$

and also

$$T^1 \cdot \varepsilon_{\pm}(k) = \frac{1}{\sqrt{2}} (1, 0, 0, 1)^T = \frac{-1}{\sqrt{2}} \frac{k}{E}, \quad T^2 \cdot \varepsilon_{\pm}(k) = \frac{\mp i}{\sqrt{2}} (1, 0, 0, 1)^T = \frac{\mp i}{\sqrt{2}} \frac{k}{E}.$$

Here we see that $T^j \varepsilon_{\pm} \propto k$, and because we have defined T^j as generators of a group which leaves k invariant, it must hold that $T^j \cdot k = 0$. Therefore, we conclude that the Taylor expansion of the exponential vanishes in higher than linear terms and we obtain

$$e^{-i(b^1 T_1 + b^2 T_2)} \varepsilon_{\pm}(k) = \varepsilon_{\pm}(k) - i(b^1 \mp ib_2) \frac{k}{\sqrt{2}E} + 0 + \dots$$

The exponentiation of J^3 is now straightforward since $\varepsilon_{\pm}$ are its eigenvectors. Overall, we obtain:

$$\varepsilon_{\pm}(\Lambda p) = \Lambda L(p) e^{\mp i\vartheta(p, \Lambda)} \left[\varepsilon_{\pm}(k) - i(b^1 \mp ib_2) \frac{k}{E} \right].$$

Now, we just apply the canonical boost to obtain

$$\varepsilon_{\pm}(\Lambda p) = e^{\mp i\vartheta(p, \Lambda)} \Lambda \cdot \varepsilon_{\pm}(p) - i(b^1 \mp ib_2) \frac{\Lambda \cdot p}{\sqrt{2}E} e^{\mp i\vartheta(p, \Lambda)}.$$

In this relation we clearly see that the first part transforms Lorentz-covariantly, while the second part spoils the covariance.

[6.] Firstly, we use the definition of the field

$$U(\Lambda) A^{\mu}(x) U^{\dagger}(\Lambda) = \sum_{h=\pm} \int d\tilde{p} \left[U(\Lambda) a(p, h) U^{\dagger}(\Lambda) \varepsilon_h^{\mu}(p) e^{-ip \cdot x} + \text{h.c.} \right].$$

Now we use the relation for the transformation of the creation/annihilation operators

$$U(\Lambda) a(p, h) U^{\dagger}(\Lambda) = a(\Lambda p, h) e^{-ih\vartheta(p, \Lambda)},$$

and therefore we have

$$\begin{aligned} U(\Lambda) A^{\mu}(x) U^{\dagger}(\Lambda) &= \sum_{h=\pm} \int d\tilde{p} \left[a(\Lambda p, h) e^{-ih\vartheta(p, \Lambda)} \varepsilon_h^{\mu}(p) e^{-ip \cdot x} + \text{h.c.} \right] \\ &= \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) e^{-ih\vartheta(p, \Lambda)} \varepsilon_h^{\mu}(\Lambda^{-1} p) e^{-ip \cdot \Lambda x} + \text{h.c.} \right], \end{aligned}$$

where we have used the Lorentz invariance of the $d\tilde{p}$ measure. Now we can use the relation derived in the previous task:

$$\begin{aligned} U(\Lambda) A(x) U^{\dagger}(\Lambda) &= \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) e^{-ih\vartheta(p, \Lambda)} e^{-ih\vartheta(p, \Lambda^{-1})} \right. \\ &\quad \left. \left(\Lambda^{-1} \cdot \varepsilon_{\pm}(p) - i(b^1 \mp ib_2) \frac{\Lambda^{-1} \cdot p}{\sqrt{2}E} \right) e^{-ip \cdot \Lambda x} + \text{h.c.} \right] \\ &= \Lambda^{-1} \cdot A(\Lambda x) - i \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) \frac{b^1 - ihb^2}{\sqrt{2}E} (\Lambda^{-1} \cdot p) e^{-ip \cdot \Lambda x} + \text{h.c.} \right] \\ &= \Lambda^{-1} \cdot A(\Lambda x) - i \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) \frac{b^1 - ihb^2}{\sqrt{2}E} (\Lambda^{-1} \cdot p) e^{-i\Lambda^{-1} p \cdot x} + \text{h.c.} \right] \end{aligned}$$

$$\begin{aligned}
 &= \Lambda^{-1} \cdot A(\Lambda x) + \partial_x \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) \frac{b^1 - i h b^2}{\sqrt{2E}} e^{-i\Lambda^{-1}p \cdot x} + \text{h.c.} \right] \\
 &= \Lambda^{-1} \cdot A(\Lambda x) + \partial_x \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) \frac{b^1 - i h b^2}{\sqrt{2E}} e^{-ip \cdot \Lambda x} + \text{h.c.} \right]
 \end{aligned}$$

Overall, we get the desired result

$$U(\Lambda) A^\mu(x) U^\dagger(\Lambda) = \Lambda_\nu{}^\mu A_\nu(\Lambda x) + \partial^\mu \alpha(\Lambda x),$$

where

$$\alpha(\Lambda x) = \sum_{h=\pm} \int d\tilde{p} \left[a(p, h) \frac{b^1(p, \Lambda^{-1}) - i h b^2(p, \Lambda^{-1})}{\sqrt{2E}} e^{-ip \cdot \Lambda x} + \text{h.c.} \right].$$